

# ARNA GHOSH

Paradigms of Intelligence, Google Research, Toronto, Canada

Email ◇ GitHub ◇ LinkedIn ◇ Scholar ◇ Twitter

Research Scientist at Google Research investigating intelligent systems by combining AI and Neuroscience.

## EDUCATION

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**McGill University & MILA, Montréal**  
Ph.D. Computer Science (Vanier Scholar)

*September 2019 - February 2025*  
CGPA: 4.0/4.0

**McGill University, Montréal**  
M.Sc. Neuroscience

*September 2017 - August 2019*  
CGPA: 4.0/4.0

**Indian Institute of Technology (IIT) Kharagpur**  
B.Tech. Electrical Engineering, Minor in Computer Science (Institute **Silver medal**)

*July 2013 - May 2017*  
CGPA: 9.26/10.0

## RELEVANT INDUSTRY EXPERIENCE

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**Research Scientist, Paradigms of Intelligence, Google Research**

*Mar 2025 - Present*

- Working on co-evolving learning algorithms and hardware architectures for efficient AI systems.

**Student Researcher, Google Deepmind**

*July 2024 - Jan 2025*

- Developing a **JAX**-based brain-inspired neural network architecture for energy-efficient AI systems.

**Research Scientist Intern, FAIR, Meta Reality Labs**

*May 2023 - Jul 2023*

- Developed unsupervised metrics of representation quality to evaluate and improve *self-supervised learning* models for time series data, specifically electromyography (EMG) data.

**AI Research Scientist Intern, Ctrl-labs, Meta Reality Labs**

*Aug 2022 - Dec 2022*

- Surgical inspection of pretrained networks to investigate learned feature space and *few-shot learning* for rapidly adapting a suitable feature space for identifying new categories.

**Scientist in Residence, Innodem Neurosciences**

*Oct 2020 - Dec 2020*

- Developed a **Tensorflow** module implementing the *Deepdream* algorithm to visualize features extracted by eye gaze-prediction neural networks and to improve interpretability for application in healthcare industry.

## PUBLICATIONS

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Full Papers:

1. M.Z. Li\*, K.K. Agrawal\*, **A. Ghosh\***, K.K. Teru, A. Santoro, G. Lajoie, B.A. Richards, *Tracing the Representation Geometry of Language Models from Pretraining to Post-training*, 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, Dec 2025.
2. **A. Ghosh\***, K.K. Agrawal\*, S. Sodhani, A. Oberman<sup>†</sup>, B.A. Richards<sup>†</sup>, *Harnessing small projectors and multiple views for efficient vision pretraining*, 38th Conference on Neural Information Processing Systems (NeurIPS), Vancouver, Dec 2024.
3. R. Chua, **A. Ghosh**, C. Kaplanis, B.A. Richards, D. Precup, *Learning Successor Features the Simple Way*, 38th Conference on Neural Information Processing Systems (NeurIPS), Vancouver, Dec 2024.
4. R. Pogodin\*, J. Cornford\*, **A. Ghosh**, G. Gidel, G. Lajoie, B.A. Richards, *Synaptic Weight Distributions Depend on the Geometry of Plasticity*, 12th International Conference on Learning Representations (ICLR), Vienna, May 2024 [**spotlight**].

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<sup>\*</sup>, <sup>†</sup> Equal contribution

5. P. Li\*, J. Cornford\*, **A. Ghosh**, B.A. Richards, *Learning better with Dale's law: a spectral perspective*, 37th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Dec 2023.
6. Z. Chorghay, V.J. Li, A. Schohl, **A. Ghosh**, E.S. Ruthazer. *The effects of the NMDAR co-agonist D-serine on the structure and function of the optic tectum*. Scientific Reports, Aug 17 2023;13(1):13383.
7. **A. Ghosh**, Y.H. Liu, G. Lajoie, K.P. Körding, B.A. Richards, *How gradient estimator variance and bias impact learning in neural networks*, 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, May 2023.
8. K.K. Agrawal\*, A.K. Mondal\*, **A. Ghosh\***, B.A. Richards,  *$\alpha$ -ReQ: Assessing representation quality by measuring eigenspectrum decay*, 36th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Dec 2022.
9. Y.H. Liu, **A. Ghosh**, B.A. Richards, E. Shea-Brown, G. Lajoie, *Beyond accuracy: generalization properties of biologically plausible temporal credit assignment rules*, 36th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Dec 2022.
10. A. Xifra-Porxas\*, **A. Ghosh\***, G.D. Mitsis, M.H. Boudrias, *Estimating brain age from structural MRI and MEG data: Insights from dimensionality reduction techniques*, NeuroImage 231, 117822, May 2021.
11. **A. Ghosh**, F. Dal Maso, M. Roig, G.D. Mitsis, M.H. Boudrias, *Unfolding the effects of acute cardiovascular exercise on neural correlates of motor learning using Convolutional Neural Networks*, Frontiers in Neuroscience 13 (2019): 1215, November 2019.
12. **A. Ghosh**, S. Singh, D. Sheet, *Simultaneous Localization and Classification of Acute Lymphoblastic Leukemic Cells in Peripheral Blood Smears Using a Deep Convolutional Network with Average Pooling Layer*, IEEE 12th International Conference on Industrial & Information Systems (ICIIS), December 2017.

#### Key Poster Presentations:

1. M. Gamba, **A. Ghosh**, K.K. Agrawal, B.A. Richards, H. Azizpour, M. Björkman *On the varied faces of model scaling in supervised and self-supervised learning*, SSL theory & practice workshop, NeurIPS, New Orleans, Dec 2023 [Oral].
2. **A. Ghosh**, K.K. Agrawal, Z. Chorghay, A.K. Mondal, B.A. Richards, *Neural Population Geometry across model scale: A tool for cross-species functional comparison of visual brain regions*, 20<sup>th</sup> Computational and Systems Neuroscience (COSYNE), Montreal, March 2023.
3. J. Cornford, **A. Ghosh**, G. Gidel, B.A. Richards, *Learning in neural networks with brain-inspired geometry*, 20<sup>th</sup> Computational and Systems Neuroscience (COSYNE), Montreal, March 2023.
4. D. Lin, R. da Silva, **A. Ghosh**, R. Tong, S. Trenholm, B.A. Richards. *Neuronal optimal stimuli synthesized with deep learning reveal functional segregations in the mouse visual cortex*. The Society for Neuroscience, 52<sup>nd</sup> Annual Meeting, San Diego, CA, November 2022.
5. **A. Ghosh\***, K.K. Agrawal\*, B.A. Richards. *Characterizing High Dimensional Representation Learning in Overparameterized Neural Networks*. Workshop on the Theory of Overparameterized Machine Learning, April 2021.
6. **A. Ghosh\***, A. Xifra-Porxas\*, G.D. Mitsis, M.H. Boudrias. *Combining structural MRI images and MEG recordings for Biological brain age prediction*, Organization of Human Brain Mapping (OHBM), Annual Meeting, June 2019. [Link](#)

## MAJOR PROJECTS

### Representation geometry in the brain and machines

Jan 2021 - Jan 2025

Doctoral thesis project, Supervisor: Dr. Blake A. Richards

- Characterizing the geometry of learned representations in artificial neural networks trained using self-supervised learning (SSL) and connecting these properties to generalization performance.
- Using population geometry to understand and improve learning dynamics in large language models (LLMs) and SSL vision models, drawing parallels to representation learning dynamics in the brain.

### Biologically-plausible credit assignment for representation learning

Sep 2019 - Jan 2025

Doctoral thesis project, Supervisor: Dr. Blake A. Richards

- Understanding the impact of imperfect credit assignment on performance of learning systems.
- Developing a biologically-plausible temporal credit assignment and a self-supervised learning algorithm for representation learning from continuous stream of visual experience.

### Deep Learning for Neuroimaging data analysis

Sep 2017 - Aug 2019

*Masters thesis project, Supervisors: Dr. Marie-Hélène Boudrias & Dr. Georgios D. Mitsis*

- Developed a deep learning-based prediction framework and a network interpretability module in **Torch** to identify exercise-induced task signatures for electroencephalography (EEG) data.
- Designed a ML-based **brain age prediction** system from magnetic resonance imaging (MRI) and magnetoencephalography (MEG) recordings in **PyTorch**.

### Brexting: Brain Texting

Dec 2017 - Aug 2018

*Intel Innovate FPGA Grand Finalists (**Silver** & **Iron** awards in regional & grand finals resp.)*

- Built a deep learning-powered brain-computer interface on an Intel FPGA board to enable real-time typing from imagined motor movements. Code available on github repo.

### Deep Learning for mitotic figure detection

Jul 2016 - May 2017

*Bachelor's Thesis Project, Supervisor: Dr. Debodoot Sheet (**Systems Society** award)*

- Built a deep learning framework to localize and identify mitotic nuclei from breast histopathological images on **Torch** and extended the idea to leukemia detection.

## TEACHING AND VOLUNTEERING EXPERIENCE

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### Student Lab Representative

Aug 2020 - Oct 2021

*Montreal Institute for Learning Algorithms (MILA)*

### Content Creator, Un/Self-supervised Learning Methods

Aug 2021

*Dr. Blake Richards & Dr. Tim Lillicrap, Neuromatch Academy: Deep Learning*

### Reviewer

*IEEE Signal Processing Magazine; IEEE Journal of Biomedical and Health Informatics; Transactions on Medical Imaging; NeurIPS 2022 (Top reviewers); ICLR 2023; NeurIPS 2023, 2024, 2025;*

### Teaching Assistant, McGill University

Jan 2019-Apr 2022

*Artificial Intelligence; Applied Machine Learning; Computational Perception; Brain-inspired AI*

### Co-organizer

*Computational and Systems Neuroscience (COSYNE) workshop on representation geometry*

Mar 2024

*Computational Cognitive Neuroscience (CCN) GAC workshop on Neural Representations*

Aug 2023

*Computational Cognitive Neuroscience (CCN) GAC workshop on Recurrent Networks*

Oct 2020

## SCHOLARSHIPS AND AWARDS

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- **Vanier** Canada Graduate Scholarship, September 2021 (\$50,000/year, 3 years)
- Healthy Brains for Healthy Lives (**HBHL**) Doctoral fellowship, September 2020 (\$15,000)
- 2<sup>nd</sup> prize in McGill **TechIdea** Pitch Competition, January 2019 (\$250)
- Healthy Brains for Healthy Lives (**HBHL**) Masters fellowship, September 2018 (\$10,000)
- McGill **Faculty of Medicine** Internal fellowship, September 2018 (\$10,000)
- Quebec Bio-Imaging Network (**QBIN**) Foreign Students Scholarship, December 2017 (\$7,000)
- **MITACS** Graduate Fellowship award 2017 for pursuing graduate studies in Canada (\$15,000)
- **Honda** Young Engineer and Scientist award, January 2016 (\$3,000)